

Feature Preparation and Experimental Design Report for SLURRY_FREE_ACID Forecasting

TSP Fertilizer Process Analytics Project

Executed notebook basis: 16 July 2026

Abstract

This report documents the fully executed `02_feature_engineering_data_preparation.ipynb`. The preparation layer now supports four direct forecasting horizons and four hypothesis-driven feature sets. It does not train a predictive model. All numerical claims, tables, and figures in this report are derived from the notebook's exported audit artifacts after a clean, top-to-bottom execution.

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1 Executive decision and scientific purpose

The preparation stage is no longer intended to maximize matrix width. Its purpose is to establish a controlled experiment that can test: (1) whether current measurements suffice; (2) whether selected histories add value; (3) whether process relationships improve prediction and interpretation; (4) whether temporal and physical evidence are complementary; and (5) how predictability degrades as warning time increases.

The direct targets are

$$y_{t+h} = \text{SLURRY_FREE_ACID}_{t+h}, \quad h \in \{1, 5, 10, 15\} \text{ minutes.}$$

Every input attached to timestamp t uses only measurements available at t or earlier. Feature-set ablation and horizon degradation are therefore first-class experimental questions for the next stage.

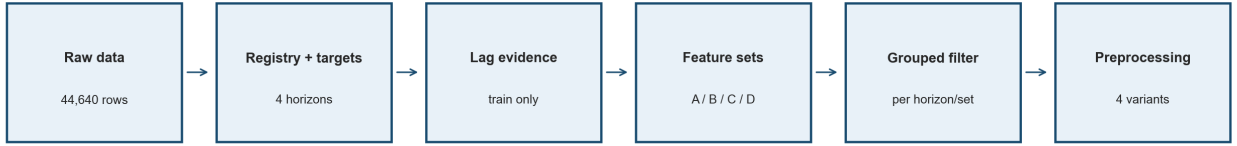


Figure 1: Executed feature-preparation pipeline.

2 Dataset and execution basis

Decision. The notebook uses one chronologically ordered one-minute dataset and excludes the target from the 21 sensor inputs.

Implementation. Timestamps are parsed as UTC and sorted before target or feature construction. The notebook was regenerated, executed in a fresh Python process, and saved with execution counts for every code cell.

Actual output. The execution completed all 11 code cells, exported 21 CSV audit tables and seven figures, and passed all readiness assertions.

Modeling meaning. The report is traceable to machine-readable artifacts rather than copied source-code intentions.

Limitation. A complete month is available, but one month cannot establish seasonal, campaign, or product-grade robustness.

Item	Executed value
Source	<code>tsp_1min.csv</code>
Raw rows	44,640
Input sensors	21
Missing target observations	308
Timestamp range	2026-01-01 00:00:00+00:00 to 2026-01-31 23:59:00+00:00
Sampling / coverage	1 minute; 31 days

Table 1: Dataset and timestamp summary.

3 Process-variable registry and intervention safety

Decision. Every source variable is registered before engineering, with a primary category, category tags, role, controllability status, measurement type, physical interpretation, interpretation status, unit-validation status, and recommendation flag.

Implementation. Candidate actuator tags are labelled `needs_supervisor_confirmation`; they are not silently promoted to controllable. `recommendation_allowed` becomes true only for `confirmed_controllable` variables.

Actual output. No variable is currently confirmed controllable and no change recommendation is authorized. Units are not supplied by the source CSV. Two tags have uncertain roles and two flow-related interpretations require process confirmation.

Modeling meaning. The registry supports grouping, explanations, scenario design, and a hard safety boundary between observed state and authorized intervention.

Limitation. Registry entries are engineering hypotheses until tag dictionaries, units, locations, and actuator authority are reviewed by the supervisor.

Primary category	Role	Variables
flow	candidate_actuator_not_confirmed	6
flow	observed_state	3
observed_state	observed_state	1
possible_quality_indicator	target	1
pressure	candidate_actuator_not_confirmed	1
pressure	observed_state	1
production	observed_state	1
production	uncertain_role	1
thermal	observed_state	7

Table 2: Variable-registry summary by category and role.

4 Multi-horizon target construction

Decision. Four direct future targets are retained to compare accuracy, warning usefulness, reaction time, and horizon degradation.

Implementation. For horizon h , the target is created with a negative h -row shift. Future target values are never features. Rows without a valid future target are removed independently for each horizon.

Actual output. Longer horizons have fewer usable rows because more observations at the end have no future label; ordinary missing target readings also remain excluded.

Modeling meaning. A highly accurate one-minute model can now be compared with less accurate but operationally earlier warnings.

Limitation. These are direct targets sharing the same input definitions; horizon-specific feature selection may still be worth testing later under nested validation.

Horizon	Target column	Usable rows	Unavailable tail	Missing targets
1 min	SLURRY_FREE_ACID_t_plus_1min	44,302	1	308
5 min	SLURRY_FREE_ACID_t_plus_5min	44,298	5	312
10 min	SLURRY_FREE_ACID_t_plus_10min	44,293	10	317
15 min	SLURRY_FREE_ACID_t_plus_15min	44,288	15	322

Table 3: Target-horizon summary after structural warm-up removal.

5 Training-only selected-lag methodology

Decision. Uniform lag generation was replaced with evidence-based selection capped at three lags per sensor.

Implementation. Candidate lags are 1, 5, 10, 15, 30, and 60 minutes. Against the one-minute future target, the raw training period only is used to calculate absolute lagged Pearson correlation, mutual information, and correlation stability over three chronological training segments. Correlation and mutual-information scores are min-max normalized, averaged, multiplied by stability, and augmented by a 0.15 response-prior bonus. The top three scores are retained.

Actual output. Sixty-three lags were selected. Twenty sensors retained 1, 5, and 10 minutes; TEMPERATURE_CUVE_ATTACHE retained 5, 10, and 15 minutes.

Modeling meaning. Selected lags represent plausible response delays supported by both association and response-time assumptions.

Limitation. This evidence is predictive, not causal. The common lag dictionary was selected against $t + 1$ for controlled feature-set comparisons and may not be optimal for every horizon.

Table 4: Selected-lag evidence and rationale.

Source variable	Selected lags	Statistical evidence (selected-lag means)	Prior hits	Final reason
DEPRESSURE_SECHEUR	1, 5, 10	$ r =0.269$; $MI=0.145$; $stability=0.945$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
FLOW_RATE_FIOUL	1, 5, 10	$ r =0.337$; $MI=0.180$; $stability=0.966$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
FLOW_RATE_GROUND_PHOSPHATE	1, 5, 10	$ r =0.292$; $MI=0.204$; $stability=0.914$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
FLOW_RATE_LIQUIDE_LAVAGE	1, 5, 10	$ r =0.265$; $MI=0.120$; $stability=0.943$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
FLOW_RATE_PHOSPHORIC_ACID_1	1, 5, 10	$ r =0.213$; $MI=0.145$; $stability=0.919$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
FLOW_RATE_PHOSPHORIC_ACID_2	1, 5, 10	$ r =0.296$; $MI=0.201$; $stability=0.933$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
FLOW_RATE_SLURRY_M3_HR	1, 5, 10	$ r =0.296$; $MI=0.178$; $stability=0.924$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
FLOW_RATE_SLURRY_M3_HR_2	1, 5, 10	$ r =0.109$; $MI=0.197$; $stability=0.294$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
FLOW_RATE_SLURRY_T_PER_HR	1, 5, 10	$ r =0.301$; $MI=0.187$; $stability=0.927$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
FLOW_RATE_VAPEUR	1, 5, 10	$ r =0.341$; $MI=0.191$; $stability=0.942$	1, 5	Top stable train-only correlation/MI score; 2/3 delays agree with response-time prior.
LEVEL_CUVE_PASSAGE	1, 5, 10	$ r =0.312$; $MI=0.143$; $stability=0.976$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
PRESSURE_VAPEUR	1, 5, 10	$ r =0.315$; $MI=0.139$; $stability=0.940$	1, 5	Top stable train-only correlation/MI score; 2/3 delays agree with response-time prior.
PRODUCTION_TSP_BALANCE	1, 5, 10	$ r =0.315$; $MI=0.199$; $stability=0.921$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
RECYCLAGE	1, 5, 10	$ r =0.344$; $MI=0.201$; $stability=0.935$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.
TEMPERATURE_AIR_CHAUD	1, 5, 10	$ r =0.307$; $MI=0.105$; $stability=0.959$	5, 10	Top stable train-only correlation/MI score; 2/3 delays agree with response-time prior.
TEMPERATURE_BRIQUE	1, 5, 10	$ r =0.297$; $MI=0.169$; $stability=0.955$	10	Top stable train-only correlation/MI score; 1/3 delays agree with response-time prior.
TEMPERATURE_CUVE_ATTACHE	5, 10, 15	$ r =0.262$; $MI=0.161$; $stability=0.941$	5, 10	Top stable train-only correlation/MI score; 2/3 delays agree with response-time prior.
TEMPERATURE_CUVE_DE_PASSAGE	1, 5, 10	$ r =0.274$; $MI=0.149$; $stability=0.957$	5, 10	Top stable train-only correlation/MI score; 2/3 delays agree with response-time prior.
TEMPERATURE_GAZ_SORTIE_SECHEUR	1, 5, 10	$ r =0.305$; $MI=0.110$; $stability=0.967$	5, 10	Top stable train-only correlation/MI score; 2/3 delays agree with response-time prior.
TEMPERATURE_SLURRY_PULVERISATEUR	1, 5, 10	$ r =0.280$; $MI=0.129$; $stability=0.959$	5, 10	Top stable train-only correlation/MI score; 2/3 delays agree with response-time prior.
TEMPERATURE_VAPEUR	1, 5, 10	$ r =0.283$; $MI=0.120$; $stability=0.947$	1, 5, 10	Top stable train-only correlation/MI score; 3/3 delays agree with response-time prior.

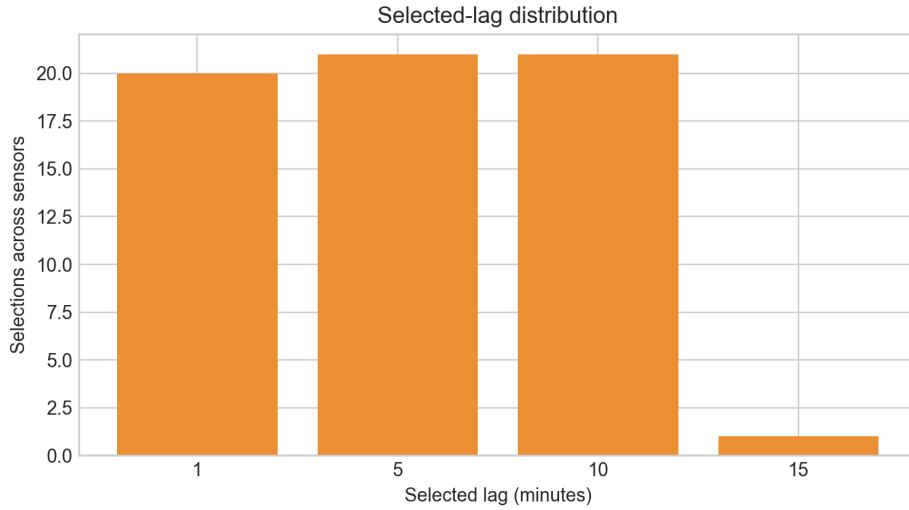


Figure 2: Distribution of the 63 selected lags.

6 Compact temporal engineering

Decision. Each sensor receives a small, consistent temporal description instead of every statistic at every window.

Implementation. The executed families are current value; three selected lags; 10- and 30-minute trailing means; 10-minute trailing standard deviation; five-minute difference; and 10-minute rolling slope. Windows are trailing and require complete history.

Actual output. The temporal candidate block contains 189 features before missingness indicators: 21 raw, 63 selected lags, 42 means, and 21 each for variability, difference, and trend.

Modeling meaning. Raw values encode present state; lags encode delayed response; means encode recent level; standard deviation encodes instability; differences encode movement; slopes encode direction.

Limitation. The same summary windows are used across sensors after lag selection. Later ablation should test whether every retained family earns its complexity.

Family	Candidates	Modeling hypothesis
<code>difference</code>	21	recent movement
<code>raw</code>	21	current process state
<code>rolling_mean</code>	42	recent operating level
<code>selected_lag</code>	63	delayed process response
<code>trend</code>	21	short-term direction
<code>variability</code>	21	short-term instability

Table 5: Temporal-feature family counts from the executed notebook.

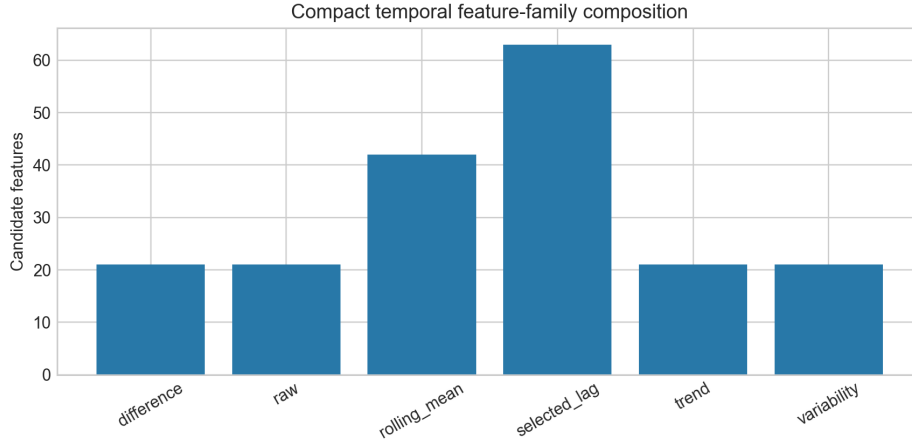


Figure 3: Compact temporal candidate composition.

7 Process-aware features and validation metadata

Decision. Physical relationships are explicit experimental features, never undocumented interactions.

Implementation. Each derived feature stores its exact formula, source tags, expected interpretation, assumptions, and validation status. Five coefficient-of-variation indicators use a 10-minute standard deviation divided by the magnitude of the 10-minute mean.

Actual output. Sixteen process-aware features were created: eleven balances, totals, ratios, gradients, or operating proxies, plus five flow-stability indicators. The total slurry volumetric flow is a confirmed m^3/h aggregation; the total acid aggregation is confirmed while its unit remains pending; eleven features remain provisional and three require mass-stream, unit, or engineering validation.

Modeling meaning. These features test whether engineering relationships generalize better and explain more clearly than sensor-by-sensor statistics.

Limitation. None is a verified physical quantity solely because it appears in this registry. In particular, pressure times temperature is a steam-condition model proxy, not energy or enthalpy.

Table 6: Complete process-aware feature registry.

Feature	Formula	Inputs	Expected interpretation	Assumptions	Status
TOTAL_ACID_FLOW	acid_1+acid_2	FLOW_RATE_PHOSPHORIC_ACID_1;FLOW_RATE_PHOSPHORIC_ACID_2	Combined phosphoric-acid feed across both lines.	The two acid lines are confirmed additive; their engineering unit still requires confirmation.	aggregation_confirmed_units_pending
ACID_FLOW_IMBALANCE	acid_1-acid_2	FLOW_RATE_PHOSPHORIC_ACID_1;FLOW_RATE_PHOSPHORIC_ACID_2	Magnitude of line imbalance.	The lines are comparable and imbalance is operationally meaningful.	provisional
ACID_TO_PHOSPHATE_RATIO	(acid_1+acid_2)/(ground_phosphate+epsilon)	FLOW_RATE_PHOSPHORIC_ACID_1;FLOW_RATE_PHOSPHORIC_ACID_2;FLOW_RATE_GROUND_PHOSPHATE	Feed-ratio proxy related to acidulation conditions.	Flow units/bases are compatible; not a stoichiometric ratio without composition data.	provisional
TOTAL_SLURRY_M3_FLOW	slurry_m3_1+slurry_m3_2	FLOW_RATE_SLURRY_M3_HR;FLOW_RATE_SLURRY_M3_HR_2	Confirmed total slurry volumetric flow in m3/h.	Both tags are confirmed additive components with the same m3/h unit.	confirmed_m3_per_h_aggregation
SLURRY_DENSITY_PROXY	slurry_t_per_h/(slurry_m3_1+slurry_m3_2+epsilon)	FLOW_RATE_SLURRY_T_PER_HR;FLOW_RATE_SLURRY_M3_HR;FLOW_RATE_SLURRY_M3_HR_2	Apparent slurry mass-per-volume proxy using total volumetric flow.	The two volumetric tags are confirmed additive in m3/h; confirm that the t/h tag represents the same combined stream.	requires_mass_stream_validation
ATTACK_PASSAGE_TEMP_GRADIENT	T_attack-T_passage	TEMPERATURE_CUVE_ATTAQUE;TEMPERATURE_CUVE_DE_PASSAGE	Thermal change between reaction vessels.	Sensors are comparable and process direction is attack to passage.	provisional
HOT_AIR_DRYER_OUTLET_GRADIENT	T_hot_air-T_dryer_outlet_gas	TEMPERATURE_AIR_CHAUD;TEMPERATURE_GAZ_SORTIE_SECHEUR	Dryer thermal-driving-force proxy.	Temperatures correspond to the same dryer train.	provisional
STEAM_SLURRY_TEMP_GRADIENT	T_steam-T_slurry	TEMPERATURE_VAPEUR;TEMPERATURE_SLURRY_PULVERISATEUR	Available temperature-gap proxy.	Steam/slurry tags belong to the relevant heat-transfer path.	provisional
STEAM_ENERGY_PROXY	steam_pressure*steam_temperature	PRESSURE_VAPEUR;TEMPERATURE_VAPEUR	Steam-condition proxy, not energy or enthalpy.	Tags refer to the same steam supply; units remain unconverted.	provisional
DRYING_INTENSITY_PROXY	fuel_flow*(T_hot_air-T_outlet)/(dryer_depression +epsilon)	FLOW_RATE_FIOUL;TEMPERATURE_AIR_CHAUD;TEMPERATURE_GAZ_SORTIE_SECHEUR;DEPRESSURE_SECHEUR	Composite burner/load/draft indicator.	Signs and units are not thermodynamically normalized; engineering review required.	requires_engineering_validation
RECYCLE_TO_PRODUCTION_RATIO	recycle/(production+epsilon)	RECYCLAGE;PRODUCTION_TSP_BALANCE	Recycle-intensity proxy.	Recycle and production tags use compatible mass/time bases.	requires_unit_validation
FLOW_RATE_PHOSPHORIC_ACID_1_CV_10	rolling_std_10(FLOW_RATE_PHOSPHORIC_ACID_1)/(rolling_mean_10(FLOW_RATE_PHOSPHORIC_ACID_1) +epsilon)	FLOW_RATE_PHOSPHORIC_ACID_1	Scale-normalized ten-minute flow instability.	Mean magnitude away from zero; extreme proxy values are clipped below.	provisional
FLOW_RATE_PHOSPHORIC_ACID_2_CV_10	rolling_std_10(FLOW_RATE_PHOSPHORIC_ACID_2)/(rolling_mean_10(FLOW_RATE_PHOSPHORIC_ACID_2) +epsilon)	FLOW_RATE_PHOSPHORIC_ACID_2	Scale-normalized ten-minute flow instability.	Mean magnitude away from zero; extreme proxy values are clipped below.	provisional
FLOW_RATE_GROUND_PHOSPHATE_CV_10	rolling_std_10(FLOW_RATE_GROUND_PHOSPHATE)/(rolling_mean_10(FLOW_RATE_GROUND_PHOSPHATE) +epsilon)	FLOW_RATE_GROUND_PHOSPHATE	Scale-normalized ten-minute flow instability.	Mean magnitude away from zero; extreme proxy values are clipped below.	provisional
FLOW_RATE_FIOUL_CV_10	rolling_std_10(FLOW_RATE_FIOUL)/(rolling_mean_10(FLOW_RATE_FIOUL) +epsilon)	FLOW_RATE_FIOUL	Scale-normalized ten-minute flow instability.	Mean magnitude away from zero; extreme proxy values are clipped below.	provisional
FLOW_RATE_VAPEUR_CV_10	rolling_std_10(FLOW_RATE_VAPEUR)/(rolling_mean_10(FLOW_RATE_VAPEUR) +epsilon)	FLOW_RATE_VAPEUR	Scale-normalized ten-minute flow instability.	Mean magnitude away from zero; extreme proxy values are clipped below.	provisional

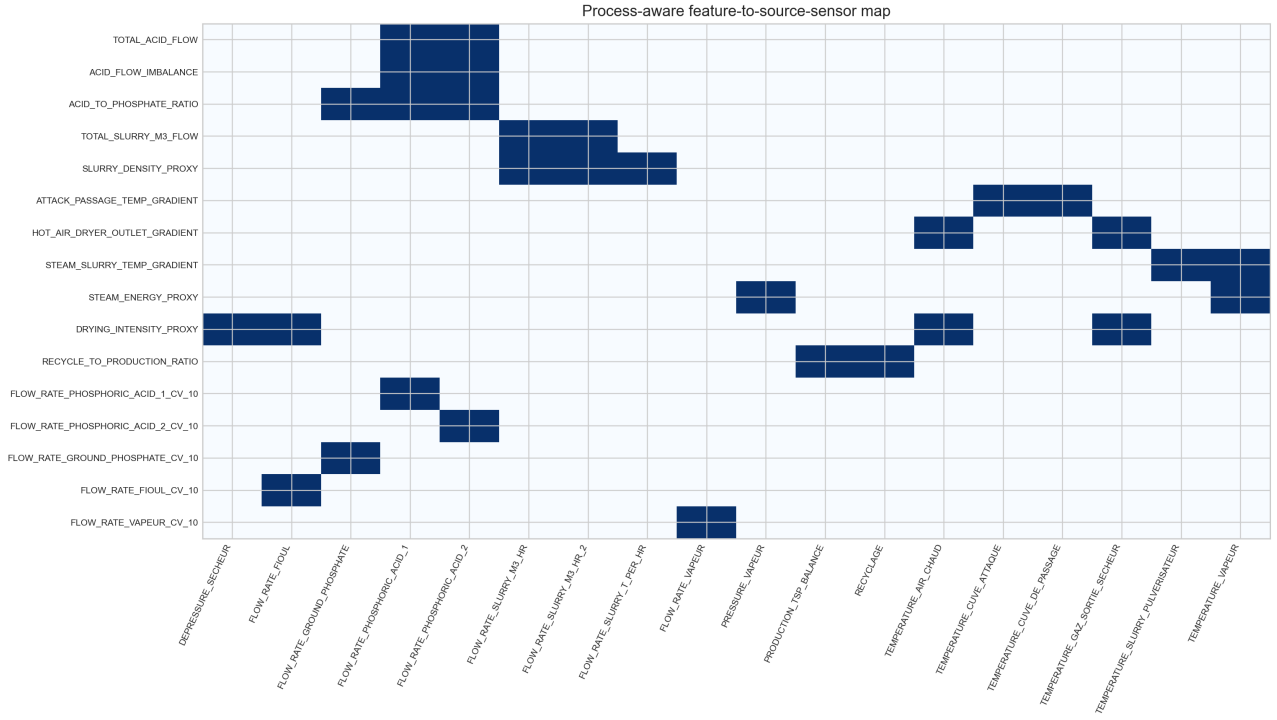


Figure 4: Process-aware features linked to their source sensors.

8 Guarded ratios, clipping, and numerical stability

Decision. Unbounded percentage changes are not generated because several sensors approach or cross zero.

Implementation. The configured epsilon is 10^{-6} . Guarded ratios are missing when $|d| \leq \epsilon$; otherwise the signed denominator receives a signed epsilon adjustment. Infinite values are replaced by missing values. Each process feature is clipped to its own 0.1st and 99.9th percentile bounds learned from the raw training period only, then applied unchanged to later periods.

Actual output. Sixteen feature-specific clipping pairs are exported in `process_clipping_bounds.csv`. The readiness audit confirms that these bounds use training data only.

Modeling meaning. The controls limit near-zero explosions and extreme leverage without using validation/test distributions.

Limitation. Quantile clipping is statistical protection, not a substitute for engineering plausibility limits. Negative source values and very large drying-proxy bounds indicate that sensor-quality and shutdown regimes need explicit review.

9 Structural warm-up versus genuine missingness

Decision. Structurally unavailable history is removed, not median-imputed.

Implementation. The maximum accepted lookback is 30 minutes (the largest selected lag is 15 minutes, but the medium rolling mean requires 30). Rows 0–29 are dropped before splitting. After this boundary, genuine gaps in all 21 sensor tags receive binary indicators in Sets B–D and median imputation fitted on training only. Set A remains strictly raw and contains no indicators.

Actual output. Exactly 30 rows were removed. The usable feature timeline begins at 2026-01-01 00:30 UTC. Twenty-one genuine-missingness indicators are available in Sets B, C, and D.

Modeling meaning. Structural absence and sensor failure now have different treatments and interpretations.

Limitation. A binary indicator identifies missingness but does not distinguish maintenance, communications failure, or process shutdown.

Policy	Rows	Reason
drop_structural_warmup	30	History does not exist for accepted lag/rolling features.
preserve_genuine_sensor_missingness	0	Post-warm-up gaps receive indicators and train-fitted median imputation.

Table 7: Warm-up and genuine-missingness policy.

10 Four feature-set hypotheses

Decision. Four named input configurations replace one universal matrix.

Implementation. Every configuration is independently filtered and preprocessed for every horizon, producing 16 experimental keys.

Actual output. Candidate widths are 21 (A), 210 (B), 58 (C), and 226 (D). Set A has no missing indicators; Sets B–D each include 21.

Modeling meaning. Direct ablation can quantify predictive gain against complexity and physical interpretability.

Limitation. Set C includes raw measurements as specified; it tests incremental process relationships rather than process features in isolation.

Set	Contents	Hypothesis
A_raw	Current sensor measurements only	The current process state alone contains sufficient predictive information.
B_temporal	Raw plus selected lags, rolling levels, variability, difference, slope, and genuine missingness indicators	Selected process histories and recent dynamics improve forecasting.
C_process	Raw plus balances, ratios, gradients, stability proxies, and genuine missingness indicators	Physically interpretable relationships improve prediction and generalization.
D_full	Accepted temporal and process-aware features plus genuine missingness indicators	Temporal and process-relationship evidence are complementary.

Table 8: Feature-set definitions and hypotheses.

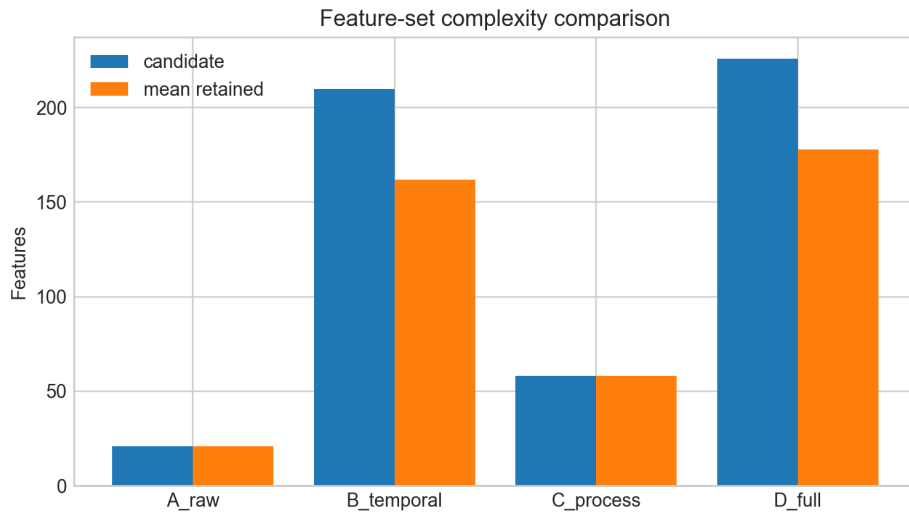


Figure 5: Candidate and mean retained feature counts by set.

11 Chronological splitting by horizon

Decision. All configurations use a 70/15/15 chronological policy without shuffling.

Implementation. Target-missing rows are removed separately by horizon before integer split boundaries are calculated.

Actual output. Split endpoints differ by minutes as the horizon increases. There is no overlap: training precedes validation, which precedes test.

Modeling meaning. Evaluation mimics forward deployment and prevents adjacent future observations from leaking backward through random allocation.

Limitation. One holdout month and highly dependent one-minute samples do not quantify seasonal uncertainty; rolling-origin validation should follow.

Horizon	Split	Rows	Start (UTC)	End (UTC)
1	train	31,011	2026-01-01 00:30:00+00:00	2026-01-22 17:17:00+00:00
1	validation	6,645	2026-01-22 17:18:00+00:00	2026-01-27 08:58:00+00:00
1	test	6,646	2026-01-27 08:59:00+00:00	2026-01-31 23:58:00+00:00
5	train	31,008	2026-01-01 00:30:00+00:00	2026-01-22 17:14:00+00:00
5	validation	6,645	2026-01-22 17:15:00+00:00	2026-01-27 08:55:00+00:00
5	test	6,645	2026-01-27 08:56:00+00:00	2026-01-31 23:54:00+00:00
10	train	31,005	2026-01-01 00:30:00+00:00	2026-01-22 17:11:00+00:00
10	validation	6,644	2026-01-22 17:12:00+00:00	2026-01-27 08:51:00+00:00
10	test	6,644	2026-01-27 08:52:00+00:00	2026-01-31 23:49:00+00:00
15	train	31,001	2026-01-01 00:30:00+00:00	2026-01-22 17:07:00+00:00
15	validation	6,643	2026-01-22 17:08:00+00:00	2026-01-27 08:46:00+00:00
15	test	6,644	2026-01-27 08:47:00+00:00	2026-01-31 23:44:00+00:00

Table 9: Exact chronological boundaries by target horizon.

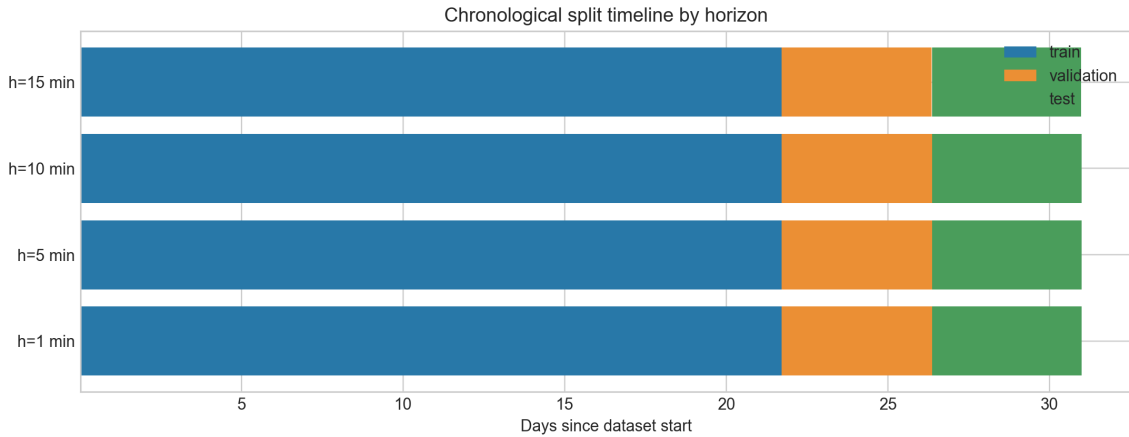


Figure 6: Chronological split timeline by horizon.

12 Grouped, interpretability-aware correlation filtering

Decision. Correlation filtering is limited to features from the same original sensor; process-aware features and missingness indicators are protected.

Implementation. The filter is fitted separately on the training matrix for each horizon and set at $|r| > 0.98$. Priority is: (1) raw value, (2) selected lag, (3) rolling mean, (4) difference, (5) slope, and (6) variability. Each removal records both features, absolute correlation, source sensor, and reason.

Actual output. Sets A and C lose no features. Sets B and D each remove 48 and retain 162 and 178 respectively at every horizon.

Modeling meaning. High correlation within a source family can be reduced while semantically distinct balances, ratios, gradients, and stability proxies remain available for ablation.

Limitation. During transitions, a raw value and lag may behave differently despite high full-period correlation. The log should be revisited for transition-focused models.

Horizon	Set	Candidate	Retained	Removed
1	A_raw	21	21	0
1	B_temporal	210	162	48
1	C_process	58	58	0
1	D_full	226	178	48
5	A_raw	21	21	0
5	B_temporal	210	162	48
5	C_process	58	58	0
5	D_full	226	178	48
10	A_raw	21	21	0
10	B_temporal	210	162	48
10	C_process	58	58	0
10	D_full	226	178	48
15	A_raw	21	21	0
15	B_temporal	210	162	48
15	C_process	58	58	0
15	D_full	226	178	48

Table 10: Grouped filtering results by horizon and feature set.

Table 11: Representative correlation-removal decisions (one-minute temporal set).

Removed	Retained	$ r $	Source sensor	Reason
PRODUCTION_TSP_BALANCE_lag_1	PRODUCTION_TSP_BALANCE	0.9924	PRODUCTION_TSP_BALANCE	Same family; retained more interpretable representative
PRODUCTION_TSP_BALANCE_lag_10	PRODUCTION_TSP_BALANCE	0.9805	PRODUCTION_TSP_BALANCE	Same family; retained more interpretable representative
PRODUCTION_TSP_BALANCE_lag_5	PRODUCTION_TSP_BALANCE	0.9881	PRODUCTION_TSP_BALANCE	Same family; retained more interpretable representative
PRODUCTION_TSP_BALANCE_roll_mean_10	PRODUCTION_TSP_BALANCE	0.9815	PRODUCTION_TSP_BALANCE	Same family; retained more interpretable representative
FLOW_RATE_PHOSPHORIC_ACID_1_lag_1	FLOW_RATE_PHOSPHORIC_ACID_1	0.9922	FLOW_RATE_PHOSPHORIC_ACID_1	Same family; retained more interpretable representative
FLOW_RATE_PHOSPHORIC_ACID_1_lag_5	FLOW_RATE_PHOSPHORIC_ACID_1	0.9860	FLOW_RATE_PHOSPHORIC_ACID_1	Same family; retained more interpretable representative
FLOW_RATE_PHOSPHORIC_ACID_2_lag_1	FLOW_RATE_PHOSPHORIC_ACID_2	0.9928	FLOW_RATE_PHOSPHORIC_ACID_2	Same family; retained more interpretable representative
FLOW_RATE_PHOSPHORIC_ACID_2_lag_5	FLOW_RATE_PHOSPHORIC_ACID_2	0.9874	FLOW_RATE_PHOSPHORIC_ACID_2	Same family; retained more interpretable representative

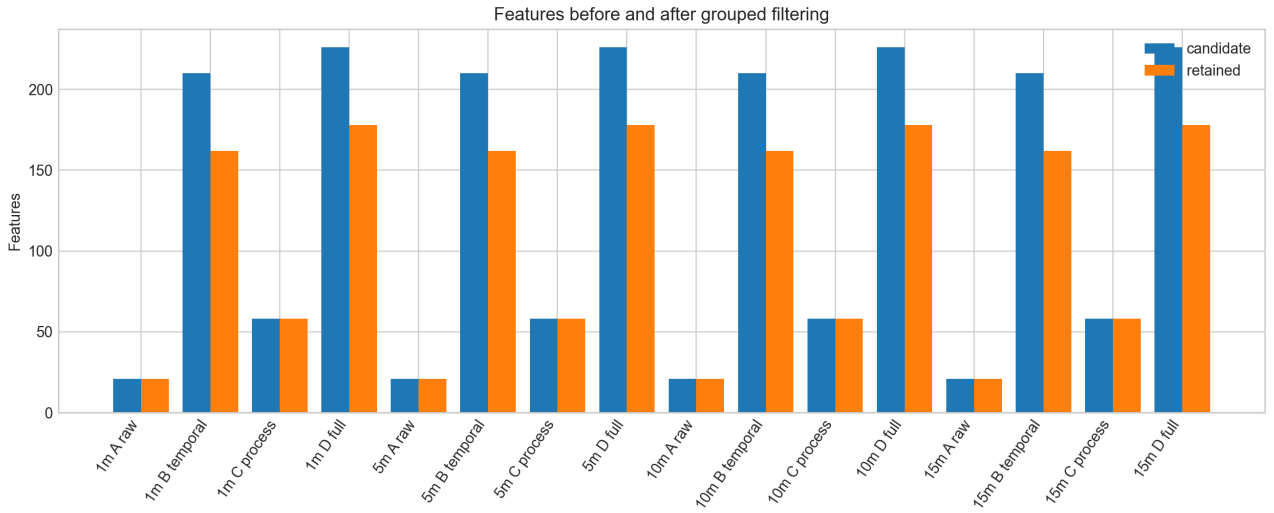


Figure 7: Features before and after grouped filtering for all 16 configurations.

13 Final dimensions and preprocessing variants

Decision. Each horizon-set pair has four preprocessing variants.

Implementation. Median imputers and any scaler are fitted on training only. Validation and test are transformed without refitting and retain identical ordered columns.

Actual output. The artifact dictionary contains 16 experiment keys and four variants per key: unscaled tree, robust, standard, and MinMax. No post-transform missing values remain.

Modeling meaning. Algorithm comparisons can use appropriate scaling without changing data partitions or feature definitions.

Limitation. MinMax values can leave the training range in future operation; this must be monitored rather than silently clipped.

Table 12: Final matrix dimensions; cells show rows $\times$ columns.

Horizon	Set	Candidate	Retained	Indicators	Train	Validation	Test
1	A_raw	21	21	0	31,011 $\times$ 21	6,645 $\times$ 21	6,646 $\times$ 21
1	B_temporal	210	162	21	31,011 $\times$ 162	6,645 $\times$ 162	6,646 $\times$ 162
1	C_process	58	58	21	31,011 $\times$ 58	6,645 $\times$ 58	6,646 $\times$ 58
1	D_full	226	178	21	31,011 $\times$ 178	6,645 $\times$ 178	6,646 $\times$ 178
5	A_raw	21	21	0	31,008 $\times$ 21	6,645 $\times$ 21	6,645 $\times$ 21
5	B_temporal	210	162	21	31,008 $\times$ 162	6,645 $\times$ 162	6,645 $\times$ 162
5	C_process	58	58	21	31,008 $\times$ 58	6,645 $\times$ 58	6,645 $\times$ 58
5	D_full	226	178	21	31,008 $\times$ 178	6,645 $\times$ 178	6,645 $\times$ 178
10	A_raw	21	21	0	31,005 $\times$ 21	6,644 $\times$ 21	6,644 $\times$ 21
10	B_temporal	210	162	21	31,005 $\times$ 162	6,644 $\times$ 162	6,644 $\times$ 162
10	C_process	58	58	21	31,005 $\times$ 58	6,644 $\times$ 58	6,644 $\times$ 58
10	D_full	226	178	21	31,005 $\times$ 178	6,644 $\times$ 178	6,644 $\times$ 178
15	A_raw	21	21	0	31,001 $\times$ 21	6,643 $\times$ 21	6,644 $\times$ 21
15	B_temporal	210	162	21	31,001 $\times$ 162	6,643 $\times$ 162	6,644 $\times$ 162
15	C_process	58	58	21	31,001 $\times$ 58	6,643 $\times$ 58	6,644 $\times$ 58
15	D_full	226	178	21	31,001 $\times$ 178	6,643 $\times$ 178	6,644 $\times$ 178

Variant	Transformation	Intended model families
tree	training-median imputation; no scaling	tree ensembles and gradient boosting
robust	training-median imputation; RobustScaler	outlier-aware linear and neural baselines
standard	training-median imputation; StandardScaler	regularized linear models, PCA, diagnostics
minmax	training-median imputation; MinMaxScaler	bounded-input neural baselines

Table 13: Preprocessing variants created for every horizon and feature set.

14 Readiness audit

The notebook’s final assertions verify execution integrity, time order, leakage boundaries, configuration completeness, fitted preprocessing, column alignment, missing-value removal, populated registries/logs, intervention safety, artifact exports, and the absence of model training. “Current or past only” is an implementation invariant: target shifts are confined to target creation; sensor lag operations use positive shifts and trailing windows.

Validation check	Result
chronological_order_preserved_for_every_horizon	PASS
four_future_targets_created	PASS
target_and_future_targets_absent_from_inputs	PASS
lag_transformations_match_positive_historical_shifts	PASS
all_four_feature_sets_prepared_for_every_horizon	PASS
preprocessor_imputation_statistics_match_training_data	PASS
train_validation_test_columns_aligned	PASS
all_prepared_matrices_have_no_missing_values	PASS
selection_cutoff_precedes_every_validation_period	PASS
process_clipping_bounds_use_selection_training_period_only	PASS
structural_warmup_removed_before_splitting	PASS
registries_and_filtering_log_populated	PASS
no_change_recommendations_authorized	PASS
no_predictive_model_trained_in_notebook	PASS
all_expected_exported_artifacts_exist	PASS

Table 14: Executed readiness checks.

15 Limitations and engineering validation requirements

The following limitations are explicit release conditions, not footnotes.

Limitation	Required validation
Only one month of one-minute data	Test on additional months, seasons, campaigns, and product grades.
Units are absent from the source CSV	Confirm every tag unit and calculation basis with process engineering.
Process-aware features are proxies	Validate stream pairing, sign conventions, and physical interpretation before operational use.
No confirmed operational thresholds	Define warning and intervention thresholds with operations and quality teams.
No variable is confirmed controllable	Authorize actuators explicitly before scenario or recommendation use.
Adjacent observations are strongly dependent	Use chronological evaluation, persistence baselines, and transition-specific diagnostics.
No predictive benchmark is trained here	Compare against persistence and simple baselines at every horizon.

Table 15: Limitations and required engineering validation.

Additional constraints are that operational warning thresholds are not defined; adjacent samples are dependent; persistence has not yet been benchmarked; tag units and stream locations remain uncertain; and product-grade or seasonal coverage cannot be inferred from January alone. Autoregressive target-history features are intentionally absent from these exogenous sets. If target history is available at deployment, it should be tested as a separate configuration to avoid obscuring the value of process inputs.

16 Implications for the next modeling stage

The next stage should not begin with the most complex algorithm. It should use the fixed experiment keys to test:

1. every model against naive persistence and simple mean/linear baselines;
2. degradation from 1 to 5, 10, and 15 minutes alongside operational warning value;
3. A versus B versus C versus D under identical chronological folds;
4. exogenous-only versus separately defined autoregressive configurations when target history is available;
5. stable operation versus transitions, shutdowns, restarts, and abnormal excursions;
6. predictive gain versus feature count, engineering validation burden, latency, and explanation quality.

Performance reporting should include average error, tail/transition error, calibration or interval coverage if uncertainty is modeled, and stability across rolling-origin folds.

17 Change log from the previous report

- Replaced the one-minute-only framing with direct 1-, 5-, 10-, and 15-minute targets.
- Replaced blanket lag and rolling generation with training-only selected lags and compact temporal families.
- Removed the obsolete 799-to-346 feature narrative; current candidates are 21, 210, 58, and 226 by feature set, with 21, 162, 58, and 178 retained.
- Replaced one universal feature matrix with 16 horizon–feature-set configurations and four preprocessing variants each.
- Replaced global correlation filtering with grouped, training-only, interpretability-prioritized filtering and a removal log.
- Replaced structural warm-up median imputation with removal of the first 30 rows, followed by indicators and train-fitted imputation for genuine gaps.
- Added variable and process-feature registries, explicit controllability safeguards, process validation statuses, training-only clipping bounds, and exported artifacts.
- Removed the recommendation to proceed immediately to sequence-model training. The correct next step is controlled benchmarking and process understanding.

18 Recommended next deliverable

The next stage should be implemented as:

03_prediction_benchmark_and_process_understanding.ipynb

It should preserve the exported experiment definitions, begin with persistence and simple baselines, and add complexity only when out-of-time evidence justifies it.

A Executed artifact inventory

The report is backed by CSV files in `reports/feature_preparation_artifacts/`: dataset summary, complete variable registry and summary, horizon summary, lag evidence and selection, temporal counts, process registry and clipping bounds, warm-up policy, feature-set definitions and membership, split boundaries, filtering summary and removal log, final dimensions, preprocessing variants, limitations, and readiness checks. Seven figures are stored in the corresponding `figures/` directory.